

# Marginal Matching Is Not Enough: Conditional Style Leakage in Factorized Generative Models

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## Abstract

A style variable in a factorized generative model is usually declared class-invariant once its aggregate distribution matches a fixed prior,  $q(z_s) \approx \mathcal{N}(0, I)$ , enforced by a *global aggregate* loss such as MMD. We argue this is the wrong criterion: it is a statement about the marginal, while class-invariance is a statement about the conditional  $q(z_s | y=k)$ , and matching the former does not imply the latter, because a well-matched marginal is compatible with strongly class-structured conditionals. We call the resulting failure *conditional style leakage*: the decoder exploits class information hidden in  $z_s$ , so class-conditional generation (sampling  $z_s$  independently of  $z_c$ ) fails silently even while reconstruction and marginal-MMD metrics look healthy.

We prove the gap formally and measure it across four unsupervised baselines (VAE, WAE-MMD,  $\beta$ -TCVAE, FactorVAE) and our own F-CS-WAE on MNIST: every model shows near-zero global MMD alongside large inter-class style separation  $\Delta_{\text{inter}}$  and linear-probe accuracy 74–100% (chance 10%). We propose *per-class style MMD*, a supervised regularizer that targets  $q(z_s | y=k)$  directly, and a post-hoc *class-conditional style prior* that repairs generation without retraining.

Instantiated in F-CS-WAE, per-class style MMD cuts  $\Delta_{\text{inter}}$  by 67–77% and raises MNIST naive-Gaussian generation self-accuracy from 15% to 70% (100% when combined with the post-hoc prior), at a clustering cost that is dataset-dependent: MNIST accuracy falls from 99.15% to 85.92%, while on CIFAR-10 the matched single-run comparison changes by only 1.5 points (81.49%  $\rightarrow$  80.00%; the 3-seed mean without the regularizer is  $80.87 \pm 0.52\%$ ). This asymmetry is itself a finding, not just a caveat: the clustering cost of removing leakage tracks how much class information the style variable was carrying in the first place, which is dataset-dependent rather than a fixed property of the regularizer. Two further results temper how far this fix can be trusted to generalize: t-SNE visualizations make the leakage visually obvious for only one of the four unsupervised baselines, despite the dependence being statistically real in all four, and the class-conditional sampling recipe that reaches 100% self-accuracy on MNIST reaches only  $\sim 70\%$  on CIFAR-10—the remedy does not transfer across datasets as cleanly as the

diagnostic that motivated it. Architectural factorization does not imply statistical independence, and supervising the dependence away is possible, but neither free nor uniformly effective.

## 1 Introduction

A factorized latent variable model is only as class-invariant as the constraint it enforces, not the constraint it intends. The standard design assigns class identity to a semantic variable  $z_c$  and within-class appearance to a style variable  $z_s$ , then regularizes  $z_s$  toward a fixed prior using a *global aggregate* objective—MMD toward  $\mathcal{N}(0, I)$  [20], total correlation penalties [1, 12], or adversarial independence constraints [2]—that operates on the marginal  $q(z_s)$ . This equates a marginal-level statistic with a conditional-level guarantee it does not imply:  $q(z_s) \approx \mathcal{N}(0, I)$  is reachable by conditionals  $q(z_s | y=k)$  that differ arbitrarily across classes, so long as they average to a Gaussian. The factorization is meant to support two goals—*clustering*, by reading class structure off  $z_c$ , and *class-conditional generation*, by fixing  $z_c$  to a class prior while sampling  $z_s$  freely—and it is the second goal that this gap quietly breaks.

**Conditional style leakage.** We call the resulting failure *conditional style leakage*. During training the decoder sees  $(z_c, z_s)$  pairs that always share class identity; if  $q(z_s | y=k)$  differs across classes, the decoder implicitly learns class-specific  $(z_c, z_s)$  interactions. At generation time, an independently sampled  $z_s$  forms a cross-class pair the decoder never saw, producing semantically inconsistent output. The failure is *silent* by construction, not by bad luck: global MMD and reconstruction SSIM are marginal- and posterior-manifold statistics respectively, and neither is positioned to detect a conditional-level dependency, so both can look healthy even when class-conditional generation is badly broken.

**Scope.** We measure this on four unsupervised baselines (VAE, WAE-MMD,  $\beta$ -TCVAE, FactorVAE) and F-CS-WAE on MNIST. All five show large inter-class style separation despite near-zero marginal discrepancy. We are careful, however, not to claim this comparison controls for architecture

or training budget (Section 4): the baselines use a lighter encoder and a shorter training schedule than F-CS-WAE, so we treat the qualitative pattern, not the ranking, as the evidence.

**Remedies and their limits.** *Per-class style MMD* directly targets each  $q(z_s | y=k)$ . The *class-conditional style prior* is a post-hoc sampling correction. Both help substantially, but neither is free or fully general: per-class style MMD trades away clustering accuracy on MNIST (not on CIFAR-10), and the class-conditional sampling recipe that perfectly repairs MNIST generation only partially repairs CIFAR-10 generation (Section 6.7).

**Case study: F-CS-WAE.** F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder) separates the latent into a hyperspherical semantic variable  $z_c \in \mathbb{S}^{d_c-1}$  aligned with class-conditional Spherical Cauchy priors and a Euclidean style variable  $z_s \in \mathbb{R}^{d_s}$ . Without per-class style MMD it reaches 99.15% clustering accuracy on MNIST but only 15% class-conditional generation self-accuracy.

**Contributions.** Our contributions are fourfold, and each answers a question the previous one raises. First, we formalize the gap between marginal matching and conditional independence and prove it is unavoidable for any aggregate divergence, not a defect of a particular loss function (Proposition 1). This raises an obvious follow-up: is the gap only a theoretical curiosity, or does it bite in practice? Second, we build a lightweight diagnostic and measure it across five model families, showing the leakage is not only real but, for three of the five, invisible to a naive 2D projection of the latent—the diagnostic is doing work a plot cannot (Section 4). Given that the failure is real, the natural question is whether it is fixable. Third, we propose per-class style MMD and show it substantially reduces, but does not eliminate, leakage, at a clustering cost that is dataset-dependent rather than uniform (Section 5). Finally, because a fix validated on one dataset is not a fix validated in general, we test whether the same remedy transfers: the class-conditional sampling recipe that is fully effective on MNIST only partially repairs CIFAR-10, a limitation invisible from the MNIST results alone (Section 6.7). We report this negative transfer result rather than omit it, since it constrains how far the proposed fix can be trusted to generalize.

## 2 Related Work

**Variational and Wasserstein auto-encoders.** VAEs [13] impose a per-example KL penalty toward a fixed Gaussian prior. WAEs [20] relax this to a global discrepancy between the aggregated posterior and the prior. MMD-based WAEs use kernel two-sample tests [6]; sliced-Wasserstein AEs [15]

use projected distances. All regularize the *marginal* of a latent variable, the setting where our gap applies.

**Disentanglement and total correlation.**  $\beta$ -VAE [9] increases the KL weight to encourage disentanglement.  $\beta$ -TCVAE [1] and FactorVAE [12] decompose the KL into a total correlation term and penalize it. InfoGAN [2] maximizes mutual information between latent codes and images. These methods penalize the joint or marginal distribution but do not explicitly target  $z_s \not\perp y$ ; Section 4 shows both still leak.

**Conditional generative models and factorized latent spaces.** Semi-supervised VAEs [14] and CVAE [19] condition generation on labels but do not study residual style independence. StyleGAN [11] factorizes structure from style via a mapping network in a class-unconditional regime. Our work addresses the label-guided setting where clustering *and* class-conditional generation are both required.

**Deep generative clustering.** DEC [22] and IDEC [7] refine cluster assignments in latent space; VaDE [10] integrates a Gaussian-mixture prior into the generative model. None diagnoses conditional style leakage, and the standard ACC/NMI/ARI protocol would not reveal the generation failure we describe.

**Hyperspherical latent variables.** S-VAE [3] replaces the Gaussian posterior with a von Mises-Fisher distribution. Spherical Cauchy VAEs [18] provide a heavier-tailed alternative with a Möbius-transform reparameterization that F-CS-WAE adopts. These geometric choices are orthogonal to the leakage problem we study.

## 3 The Conditional Style Leakage Problem

### 3.1 Setting and Notation

Let  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  be a labeled dataset with  $K$  classes. A factorized model encodes each input into a semantic variable  $z_c$  and a style variable  $z_s$ . The style variable is regularized by

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

where  $D$  is any aggregate divergence and  $q(z_s) = \mathbb{E}_y[q(z_s | y)]$ . Class-conditional generation samples  $z_s \sim \mathcal{N}(0, I)$  independently of  $z_c$ .

### 3.2 Formal Gap

**Proposition 1** (Marginal matching does not imply conditional independence). *Suppose  $q(z_s | y=k) = \mathcal{N}(\mu_k, \Sigma_k)$  for  $k = 1, \dots, K$  with  $p(y) = \text{Uniform}(\{1, \dots, K\})$ , so  $q(z_s) = \frac{1}{K} \sum_k \mathcal{N}(\mu_k, \Sigma_k)$ . Matching  $q(z_s) \approx \mathcal{N}(0, I)$*

via any aggregate divergence is compatible with arbitrary class structure in the conditionals, provided the mixture averages to a Gaussian:  $q(z_s) \approx \mathcal{N}(0, I)$  does not imply  $q(z_s | y=k) \approx \mathcal{N}(0, I)$  for any  $k$ , nor  $z_s \perp y$ .

*Proof.* A mixture of Gaussians can approximate any distribution [4], including  $\mathcal{N}(0, I)$ , with components of arbitrary mean and covariance. Concretely, take  $K = 2$ ,  $\mu_1 = -\mu_2 = v \in \mathbb{R}^{d_s}$ ,  $\Sigma_1 = \Sigma_2 = I - \frac{1}{2}vv^\top$ : the mixture mean is 0 and its covariance approaches  $I$  as  $\|v\|$  grows moderately, while the inter-class separation  $2\|v\|$  is unbounded. A full construction with  $\Delta_{\text{inter}} \rightarrow \infty$  under an exactly matched marginal is given in Appendix D.  $\square$

**Remark 1.** A regularizer minimizing  $D(q(z_s), \mathcal{N}(0, I))$  has zero gradient incentive to reduce inter-class separation, since the mixture can match the target while conditionals remain class-structured.

### 3.3 The Decoder Amplification Mechanism

During training the decoder receives  $(z_c, z_s)$  pairs from the same image; both encode  $y$  implicitly,  $z_s$  because  $q(z_s | y=k)$  is not class-invariant. At generation time  $z_c \sim p(z_c | y=k)$  encodes a specific class while  $z_s \sim \mathcal{N}(0, I)$  averages over all classes, producing a cross-class pair never seen in training. Severity depends on how class-structured  $q(z_s | y)$  becomes: visually uniform classes (MNIST digits) force more class information into  $z_s$ , amplifying the failure; high-diversity classes (CIFAR-10) leave more class-neutral variation in  $z_s$ .

### 3.4 Diagnostic Metrics

**Global MMD**  $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$ : near zero signals only that marginal matching succeeded.

**Inter-class style separation**  $\Delta_{\text{inter}} = \frac{1}{\binom{K}{2}} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$ ,  $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$ : large values indicate class-dependent style means despite small global MMD.

**Linear probing accuracy**  $\text{LP}(z_s \rightarrow y)$  : accuracy of a linear classifier predicting  $y$  from  $z_s$ ; well above chance  $1/K$  confirms recoverable class information.

**HSIC**( $z_s, y$ ) : kernel-based dependence measure [5]; non-zero indicates  $z_s \not\perp y$  regardless of marginal shape.

A model with near-zero global MMD but large  $\Delta_{\text{inter}}$ , elevated LP, or non-zero HSIC is exhibiting conditional style leakage.

**Table 1:** Conditional style leakage across five model families on MNIST. Despite near-zero global MMD in every case, inter-class style separation  $\Delta_{\text{inter}}$  is large and linear probing (LP) recovers class information well above the 10% chance rate. The last row is F-CS-WAE with the proposed per-class style MMD (Section 5): leakage drops sharply but is not eliminated. Self-ACC uses naive  $z_s \sim \mathcal{N}(0, I)$  sampling.

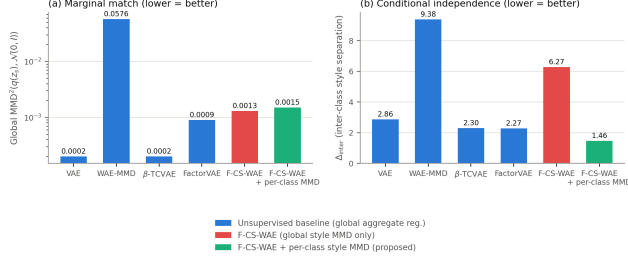
| Model              | MMD ↓  | $\Delta_{\text{inter}}$ ↓ | LP ↓   | Self-ACC ↑ |
|--------------------|--------|---------------------------|--------|------------|
| VAE                | 0.0002 | 2.86                      | 87.4%  | N/A        |
| WAE-MMD            | 0.0576 | 9.38                      | 93.0%  | N/A        |
| $\beta$ -TCVAE     | 0.0002 | 2.30                      | 73.9%  | N/A        |
| FactorVAE          | 0.0009 | 2.27                      | 75.9%  | N/A        |
| F-CS-WAE           | 0.0013 | 6.27                      | 100.0% | 0.15       |
| F-CS-WAE+per-class | 0.0015 | 1.46                      | 45.6%  | 0.70       |

## 4 Cross-Model Validation

To test whether conditional style leakage is general rather than an artifact of F-CS-WAE’s design, we apply the diagnostic of Section 3.4 to four unsupervised baselines (VAE, WAE-MMD,  $\beta$ -TCVAE, FactorVAE) and F-CS-WAE on MNIST. These four models are purpose-built for this diagnostic only, with a lightweight 3-layer convolutional encoder trained for 100 epochs; they are *not* the same checkpoints as the like-named VAE/WAE-MMD entries used later for CIFAR-10 clustering comparison (Section 6.2), which follow a different protocol (Appendix B). F-CS-WAE, by contrast, uses its full ResNet-18 encoder trained for 300 epochs under the curriculum of Section 5, so architecture and training budget are not matched across the five rows of Table 1 and a reader should not treat the  $\Delta_{\text{inter}}$  ranking between F-CS-WAE and the baselines as a controlled comparison. What the mismatch does not undermine is the *within-model* evidence: for every row individually, global MMD is decoupled from  $\Delta_{\text{inter}}$  and LP regardless of that row’s encoder capacity or training length, which is the comparison the diagnostic actually depends on.

All four baselines show LP far above chance (74–93%) despite near-zero global MMD. F-CS-WAE shows the most extreme leakage ( $\Delta_{\text{inter}} = 6.27$ , LP= 100%), consistent with MNIST’s low intra-class diversity forcing more class information into  $z_s$ . WAE-MMD has the largest global MMD among baselines (0.0576, 30–300× the others), suggesting its own regularizer had not converged, and correspondingly the largest baseline  $\Delta_{\text{inter}}$  (9.38) and LP (93%).  $\beta$ -TCVAE and FactorVAE show the smallest  $\Delta_{\text{inter}}$  (2.30, 2.27), but LP stays far above chance (74–76%): penalizing joint/marginal statistics reduces but does not eliminate the dependence. Per-class style MMD cuts F-CS-WAE’s  $\Delta_{\text{inter}}$  by 77% ( $6.27 \rightarrow 1.46$ ) and LP by 54 points ( $100\% \rightarrow 45.6\%$ )—still far above chance, and higher than three of the four unsupervised baselines. Naive self-accuracy rises correspondingly from 0.15 to 0.70.

Marginal matching vs. conditional independence across five model families (MNIST)



**Figure 1:** Global MMD (log scale, panel a) vs.  $\Delta_{\text{inter}}$  (panel b) across the six rows of Table 1. Marginal matching quality is not predictive of conditional independence.

**Is the leakage visible without the diagnostic?** A natural question is whether Table 1 is telling us something a simple visualization would already show. Figure 2 projects  $z_s$  (2000 test points) with t-SNE, colored by class, for all six rows. *Read this figure the opposite way from a typical clustering figure:*  $z_s$  is supposed to be class-invariant, so a well-mixed, color-blended cloud is the desired (good) outcome and tight, color-separated clusters are the failure mode (high leakage), consistent with the direction of  $\Delta_{\text{inter}}$  in Table 1. By that reading, the answer is mixed: WAE-MMD and F-CS-WAE (no per-class MMD)—the two rows with the *largest*  $\Delta_{\text{inter}}$ , i.e. the two worst rows in Table 1—show clearly separated clusters, and F-CS-WAE + per-class MMD (lower  $\Delta_{\text{inter}}$ , i.e. improved) shows visibly less separated, more mixed clusters, matching the table. VAE,  $\beta$ -TCVAE, and FactorVAE show *no* visually obvious clustering at all, despite LP accuracies of 74–87% in the same latent space. This is itself evidence for our methodological point: for three of the five baseline-scale models, marginal inspection or a naive 2D projection would miss the leakage entirely, and only the quantitative diagnostic (a linear probe trained on the full 64-dimensional  $z_s$ ) recovers it.

## 5 Method

### 5.1 Problem Setting: Label-Guided Generative Clustering

Given labeled images  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  with  $K$  classes, we want (1) reconstruction that preserves visual content, (2) clustering that recovers class structure from semantic codes without using labels at evaluation time, and (3) class-conditional generation that produces class-consistent images from class priors.

### 5.2 Factorized Architecture: F-CS-WAE

A shared ResNet-18 backbone extracts  $h = f_\phi(x)$ , followed by a 256-unit FC layer with SiLU, then two heads:

$$E_\phi(x) = (\tilde{\mu}_c, r_c, \mu_s, \log \sigma_s^2). \quad (2)$$

The semantic direction is  $\ell_2$ -normalized with bounded concentration:

$$\mu_c = \frac{\tilde{\mu}_c}{\|\tilde{\mu}_c\|_2}, \quad \rho_c = \sigma(r_c)(1 - \varepsilon), \quad (3)$$

sampled via the Spherical Cauchy Möbius reparameterization [18]:

$$\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1}), \quad z_c = T_{\mu_c, \rho_c}(\eta_c). \quad (4)$$

The style latent uses a Gaussian reparameterization  $z_s = \mu_s + \sigma_s \odot \varepsilon_s$ ,  $\varepsilon_s \sim \mathcal{N}(0, I)$ . Both are concatenated and decoded,  $\hat{x} = D_\theta([z_c; z_s])$ , by a residual upsampling decoder (512→256→128→64 channels,  $4 \times 4 \rightarrow 32 \times 32$ ). In all experiments  $d_c = 64$ ,  $d_s = 128$  (Appendix A).

### 5.3 Class-Conditional Semantic Prior

For each class  $k$ , F-CS-WAE maintains a center  $m_k \in \mathbb{S}^{d_c-1}$  with prior  $p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p)$ ,  $\rho_p = 0.7$ , updated by EMA from batch class-mean directions once per epoch:

$$m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau) \bar{\mu}_c^{(k)}), \quad \tau = 0.95. \quad (5)$$

### 5.4 Loss Terms

$$\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x}), \quad (6)$$

$$\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}^2(Q_k, P_k), \quad (7)$$

$$\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}, \{z_{c,r_j}^p\}), \quad (8)$$

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}}(\{z_s^i\}, \{\varepsilon_j\}), \quad (9)$$

$$\mathcal{L}_{\text{style-cl}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i: y_i=k\}, \{\varepsilon_j: |y_i=k|\}), \quad (10)$$

$$\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y), \quad (11)$$

where  $Q_k = \{z_c^i: y_i = k\}$ ,  $P_k \sim p(z_c | y=k)$ , and  $\lambda_1=1.0$ ,  $\lambda_{\text{LPIPS}}=0.1$ . Eq. 9 aligns the marginal style distribution but, as shown in Section 3, does not prevent conditional leakage; Eq. 10 is our proposed fix, targeting each conditional  $q(z_s | y=k)$  directly. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \delta(t)\mathcal{L}_{\text{style-cl}} + \eta(t)\mathcal{L}_{\text{cls}}, \quad (12)$$

with time-varying coefficients over a four-phase, 300-epoch curriculum: (A, 0–49) reconstruction only; (B, 50–99) ramp  $\mathcal{L}_{\text{style}}$ ,  $\mathcal{L}_{\text{style-cl}}$ ,  $\mathcal{L}_{\text{cls}}$ , so per-class style regularization begins before the decoder can learn to exploit class structure in  $z_s$ ; (C, 100–199) ramp  $\mathcal{L}_{\text{class}}$ ,  $\mathcal{L}_{\text{agg}}$  to half weight; (D, 200–299) ramp both to full weight (exact schedule in Appendix C).



**Figure 2:** t-SNE of the style latent  $z_s$ , colored by MNIST digit class, for all six rows of Table 1. Unlike a typical clustering figure, *color-separated clusters here mean worse* (class information leaking into  $z_s$ ): separation is visually obvious only for WAE-MMD and F-CS-WAE (no per-class MMD)—the two rows with the *largest*  $\Delta_{\text{inter}}$  in Table 1—and visibly reduced (improved) after adding per-class style MMD. For VAE,  $\beta$ -TCVAE, and FactorVAE the leakage is statistically real (LP 74–87%) but not visually apparent in this 2D projection.

## 5.5 Inference and Generation

**Clustering.** K-means on deterministic  $\mu_c$ , Hungarian-matched to compute ACC/NMI/ARI; labels are not used.

**Naive prior sampling.**  $z_c \sim p(z_c | y=k)$ ,  $z_s \sim \mathcal{N}(0, I)$ . Reliable only when per-class style MMD is used.

**Class-conditional style prior (post-hoc).** Without per-class style MMD, estimate  $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$ ,  $\bar{\sigma}_s^{2(k)} = \text{Var}[\mu_s | y=k]$  from encoded training data, then sample  $z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)}))$  with  $\tau = 0.25$ . On MNIST this restores generation self-accuracy from 15% to 100% on an existing checkpoint with no retraining (Section 6.6); Section 6.7 shows this recipe is only partially effective on CIFAR-10.

## 6 Experiments

### 6.1 Setup

**Datasets.** CIFAR-10 [16] ( $3 \times 32 \times 32$ ,  $K=10$ ) is the primary clustering/generation benchmark. MNIST [17] and Fashion-MNIST [21] ( $1 \times 32 \times 32$ ,  $K=10$ ) are used for diagnostic analysis.

**Baselines.** On CIFAR-10 (Section 6.2): ResNetAE, VAE, WAE-MMD, VaDE, single-seed under the unified lightweight-CNN protocol described in Appendix B. On MNIST (Section 6.3): the single-latent CS-WAE (one Spherical Cauchy semantic latent, no style variable), which we did not train on CIFAR-10 and so do not include in the CIFAR-10 comparison.

**Metrics.** Clustering: ACC, NMI, ARI. Reconstruction: SSIM, PSNR, LPIPS [23]. Sample quality: FID [8] (10,000 generated vs. 10,000 real). Generation self-accuracy: fraction of generated images the trained auxiliary classifier assigns to the sampled class. Leakage: global MMD,  $\Delta_{\text{inter}}$ , LP, HSIC.

**Table 2:** CIFAR-10 F-CS-WAE results, three random seeds (all real).

| Seed        | ACC           | NMI           | ARI           | FID          |
|-------------|---------------|---------------|---------------|--------------|
| 0           | 0.8043        | 0.6426        | 0.6259        | 83.83        |
| 1           | 0.8160        | 0.6607        | 0.6452        | 83.45        |
| 2           | 0.8059        | 0.6455        | 0.6264        | 81.85        |
| <b>Mean</b> | <b>0.8087</b> | <b>0.6496</b> | <b>0.6325</b> | <b>83.04</b> |
| Std         | 0.0052        | 0.0079        | 0.0090        | 0.86         |

**Table 3:** CIFAR-10 vs. baselines. F-CS-WAE is a 3-seed mean (Table 2); all other rows are a single seed-0 run (no multi-seed baseline data available). Leakage diagnostics do not apply to these single-latent baselines.

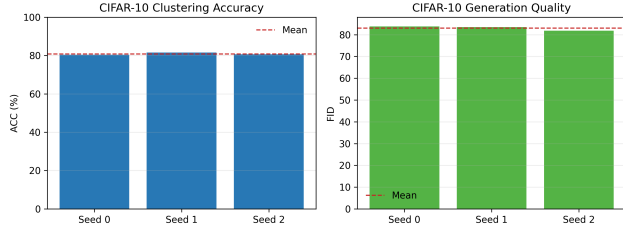
| Method                   | ACC $\uparrow$ | NMI $\uparrow$ | ARI $\uparrow$ | FID $\downarrow$ |
|--------------------------|----------------|----------------|----------------|------------------|
| <b>F-CS-WAE (3-seed)</b> | <b>0.8087</b>  | <b>0.6496</b>  | <b>0.6325</b>  | <b>83.04</b>     |
| ResNetAE                 | 0.2202         | 0.1017         | 0.0493         | 121.0            |
| VAE                      | 0.2220         | 0.0955         | 0.0511         | 180.0            |
| WAE-MMD                  | 0.2099         | 0.0979         | 0.0469         | 235.4            |
| VaDE                     | 0.1475         | 0.0427         | 0.0151         | 475.9            |

### 6.2 CIFAR-10 Main Results

Clustering is stable across seeds (ACC std 0.52 points). Table 3 compares against baselines, all measured on a single seed (seed 0) under the same unified CNN protocol; we do not have multi-seed baseline runs, so we report the single measurement without inflating it into a synthetic mean.

### 6.3 MNIST: Extreme Conditional Style Leakage

F-CS-WAE reaches 99.15% clustering accuracy and SSIM 0.983, yet class-conditional generation self-accuracy under naive sampling is only 15%: near-perfect scores on every standard metric coexist with catastrophic generation failure. Global MMD to  $\mathcal{N}(0, I)$  is 0.0013;  $\Delta_{\text{inter}} = 6.27$ ; a linear probe on  $z_s$  reaches 100.0% (chance 10%).



**Figure 3:** CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low.

**Table 4:** MNIST comparison, single seed (seed 0; no multi-seed MNIST data available). CS-WAE: single-latent Spherical Cauchy WAE.  $\Delta_{\text{inter}}$ /LP measured on the F-CS-WAE checkpoint.

| Method   | ACC    | NMI    | ARI    | FID   | $\Delta_{\text{inter}}$ |
|----------|--------|--------|--------|-------|-------------------------|
| CS-WAE   | 0.8735 | 0.8863 | 0.8411 | 26.95 | —                       |
| F-CS-WAE | 0.9915 | 0.9758 | 0.9813 | 73.99 | 6.27                    |

#### 6.4 Fashion-MNIST

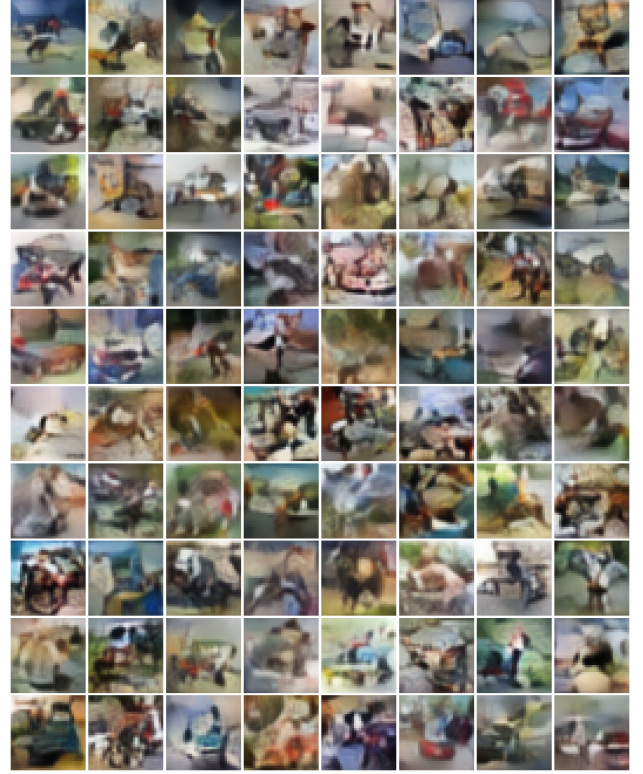
Clustering accuracy (93.4%) sits between MNIST (99.2%) and CIFAR-10 ( $\sim 81\%$ ), consistent with Fashion-MNIST’s intermediate intra-class visual diversity (Section 3.3).

#### 6.5 Cross-Dataset Leakage Comparison

Section 3.3 predicts leakage severity should track intra-class visual diversity: MNIST (least diverse)  $>$  Fashion-MNIST  $>$  CIFAR-10 (most diverse). We test this directly by running the leakage diagnostic (Section 3.4) on the  $\delta=0$  F-CS-WAE checkpoint for all three datasets.

$\Delta_{\text{inter}}$  and LP support the hypothesis cleanly: MNIST (6.27, 100.0%)  $>$  Fashion-MNIST (4.86, 97.7%)  $>$  CIFAR-10 (4.19, 89.9%). Naïve-Gaussian generation self-accuracy does not: MNIST 0.15, Fashion-MNIST 0.17, CIFAR-10 0.12—CIFAR-10 is lowest, not highest. The two families of metrics differ in what they are computed on, and that difference explains the disagreement rather than undermining either measurement.  $\Delta_{\text{inter}}$  and LP operate directly on  $z_s$  and therefore isolate the dependency we care about. Self-accuracy is measured on the generated *image*, mediated by an auxiliary classifier, so it additionally inherits how confusable the real classes already are under blurry generation—a dataset-difficulty effect with nothing to do with conditional style leakage, and one that CIFAR-10 has on top of leakage. We see no principled reason to discard the metric that disagrees, and doing so would let us report a cleaner story than the data supports; the disagreement instead tells us that generation self-accuracy, on its own, is not a reliable cross-dataset proxy for leakage severity, even though it remains the right metric for asking whether a specific checkpoint’s generation is repaired (Section 6.6).

F-CS-WAE Class-Conditional Prior Samples: rows = class, columns = style samples



**Figure 4:** CIFAR-10 class-conditional prior samples (one row per class). FID 83 places these in the recognizable-but-blurry regime typical of L1/LPIPS-trained decoders without an adversarial loss, well short of photorealistic (Section 7).

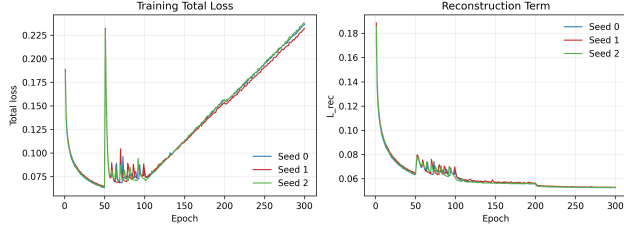
#### 6.6 Representation-Aware Sampling Strategies: MNIST

Any class-conditional strategy restores perfect self-accuracy on MNIST. The class-conditional diagonal Gaussian with  $\tau=0.25$  is a true parametric sampler with perfect class consistency and better diversity than the class-mean point estimate.

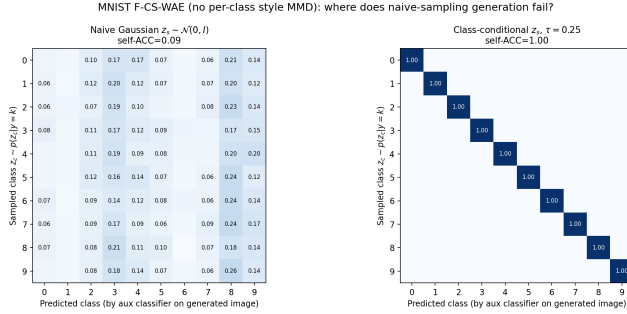
#### 6.7 Representation-Aware Sampling Strategies: CIFAR-10

Table 6 and Figure 10 show every class-conditional strategy reaching 100% self-accuracy on MNIST. We test whether the same recipe transfers to CIFAR-10, using the  $\delta=0$  CIFAR-10 checkpoint (Table 2)—a measurement not present in earlier drafts of this work.

The gap is large: the  $\tau=0.25$  recipe that is perfect on MNIST reaches only 68.1% on CIFAR-10, and the qualitative grid (Figure 12) shows why the number alone overstates the fix—samples within a class collapse toward a single low-diversity image, and several classes are not clearly identifiable by eye. Only the empirical style bank, which resamples



**Figure 5:** CIFAR-10 training curves. Phase transitions at epochs 50, 100, 200.



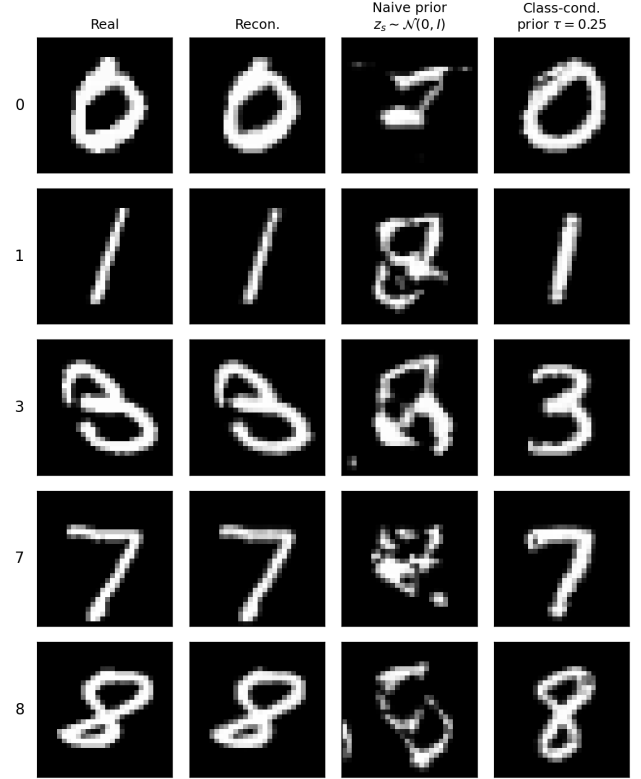
**Figure 6:** Confusion between the sampled class  $y=k$  and the class the auxiliary classifier assigns to the generated image. Naive Gaussian sampling (left) does not fail randomly—probability mass collapses onto a handful of “attractor” digits (notably 8) regardless of the intended class. The class-conditional prior (right,  $\tau=0.25$ ) is a clean diagonal.

real encoded style codes rather than fitting a per-class Gaussian, comes close to MNIST-level self-accuracy (95.9%), suggesting CIFAR-10’s per-class style distribution is not well approximated by a unimodal Gaussian, unlike MNIST’s. We treat this as a genuine limitation of the proposed remedy rather than omit it: the diagnostic (Section 6.5) generalizes across datasets more cleanly than the fix does.

## 6.8 Per-Class Style MMD: Clustering-Leakage Trade-off

On CIFAR-10, per-class style MMD is close to free: ACC changes by only 1.5 points ( $0.8149 \rightarrow 0.8000$ , within normal seed variance; Table 2),  $\Delta_{\text{inter}}$  drops 67%, and naive-sampling FID improves ( $83.59 \rightarrow 74.42$ ). On MNIST the leakage reduction is larger in relative terms (77%) but clustering accuracy drops substantially ( $99.15\% \rightarrow 85.92\%$ ). We interpret this as evidence that on MNIST—where classes are visually uniform—the style variable had been carrying class-discriminative information the semantic variable did not fully capture; suppressing it via  $\mathcal{L}_{\text{style-cl}}$  removes a source of leakage but also removes a signal the K-means-on- $\mu_c$  pipeline was implicitly benefiting from.

Per-class style MMD alone does not close the gap: naive-sampling self-accuracy on the  $\delta=1$  MNIST checkpoint is 70% (Table 1), not 100%. Reaching 100% requires *combin-*



**Figure 7:** Real image, posterior reconstruction, naive-prior sample, and class-conditional-prior sample, for five representative digits. Reconstructions are visually indistinguishable from the real image (SSIM 0.983); naive-prior samples are legible digits, but frequently the *wrong* digit (e.g. row “0”  $\rightarrow$  looks like a “3”; row “1”  $\rightarrow$  looks like a “9”); the class-conditional prior restores the intended identity.

**Table 5:** Fashion-MNIST F-CS-WAE (baseline,  $\delta=0$ ), single seed.

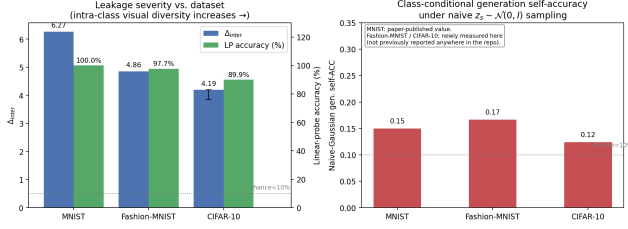
| Seed | ACC    | NMI    | ARI    | FID   |
|------|--------|--------|--------|-------|
| 0    | 0.9341 | 0.8672 | 0.8633 | 72.37 |

ing both remedies—applying the post-hoc class-conditional prior ( $\tau=0.25$ ) on top of the  $\delta=1$  checkpoint, rather than either remedy alone—which we verified directly on this checkpoint (naive 0.70  $\rightarrow$  class-conditional 1.00, same protocol as Table 6). We read this as showing the two remedies are complementary rather than redundant: per-class style MMD reduces how much class information  $z_s$  carries, and the post-hoc prior corrects for what remains.

## 7 Discussion

**Why CIFAR-10 is milder than MNIST, with a caveat.** CIFAR-10’s higher intra-class visual diversity weakens conditional style dependence, and  $\Delta_{\text{inter}}$ /LP confirm this cleanly (Section 6.5). But Section 6.7 shows the practical remedy does not inherit this cleanly: the class-conditional Gaussian prior that perfectly fixes MNIST generation only par-

Cross-dataset test of the paper’s own hypothesis (Discussion, §7): does leakage track intra-class diversity?



**Figure 8:** Left:  $\Delta_{\text{inter}}$  and linear-probe accuracy across datasets, ordered by increasing intra-class visual diversity—both decrease monotonically, matching the hypothesis. Right: naive-Gaussian generation self-accuracy does *not* follow the same order (CIFAR-10 is lowest, not highest); we attribute this to CIFAR-10 classification being intrinsically harder for the auxiliary classifier under blurry samples, confounding self-accuracy as a leakage-severity proxy (Section 7).

**Table 6:** MNIST generation under five  $z_s$  sampling strategies on the  $\delta=0$  checkpoint;  $z_c$  always from the class prior.

| Style sampling $z_s$     | Self-ACC     | Conf.        | Div.         |
|--------------------------|--------------|--------------|--------------|
| Global Gaussian          | 0.150        | 0.934        | 0.151        |
| Class mean               | 1.000        | 1.000        | 0.132        |
| Class diag., $\tau=0.50$ | 1.000        | 0.990        | 0.139        |
| Class diag., $\tau=0.25$ | <b>1.000</b> | <b>1.000</b> | 0.135        |
| Empirical style bank     | 1.000        | 1.000        | <b>0.182</b> |

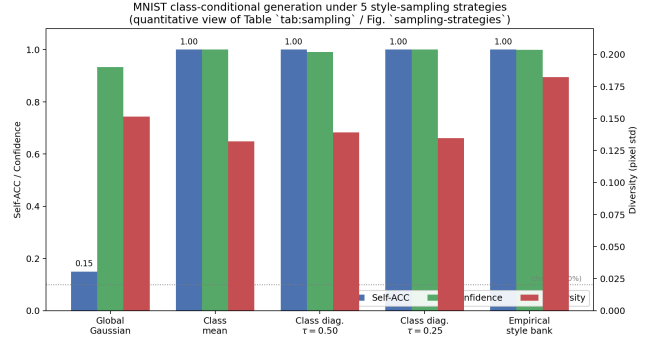
tially fixes CIFAR-10 generation, and CIFAR-10 samples remain visibly blurry (FID  $\sim 83$ – $84$ , well above photorealistic ranges). We avoid describing CIFAR-10 naive-sampling output as “visually plausible”; at this FID the more accurate description is “recognizable but not sharp,” and under the  $\tau=0.25$  class-conditional prior several classes are not reliably identifiable by eye despite scoring well on the auxiliary classifier (Figure 12).

**Global vs. conditional regularization.** Global aggregate matching is standard in WAEs and disentanglement models; our results show it is insufficient whenever the style variable must be class-invariant for generation. Per-class style MMD is a supervised analogue of total correlation penalties, targeting the single dependency  $z_s \not\perp y$  directly.

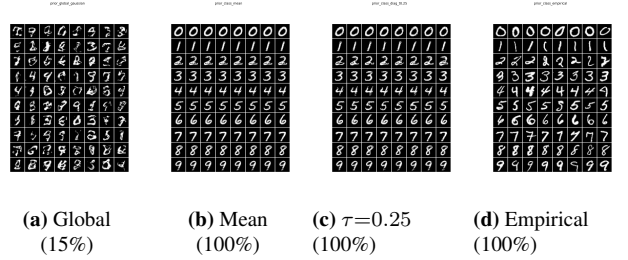
**The reconstruction–generation gap.** The MNIST checkpoint reaches SSIM 0.983 yet FID 73.99 under naive prior sampling (Figure 7). Reconstruction quality measures fidelity on the posterior manifold, not the prior; evaluation of generative clustering models should report both.

## 7.1 Limitations

We list what this paper does *not* establish, rather than leave it implicit. (1) Baseline comparisons (Tables 3, 4) and Fashion-MNIST (Table 5) are single-seed; we do not report multi-seed



**Figure 9:** Table 6 plotted directly: self-accuracy jumps from near-chance to 100% the moment  $z_s$  is conditioned on class in any way, while diversity (right axis) is lowest for the class-mean point estimate and highest for the empirical style bank—the diversity/consistency trade-off that motivates recommending  $\tau=0.25$  over the mean as a true sampler.



**Figure 10:** MNIST generation under representation-aware style sampling. Global Gaussian produces shapeless or wrong-class images; all class-conditional strategies restore correct digit identity.

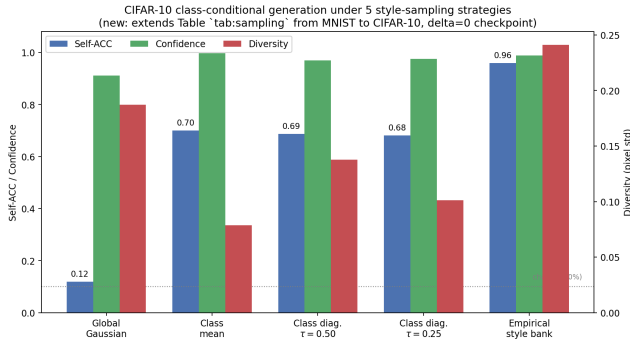
variance we have not measured. (2) The clustering-leakage trade-off (Figure 13) is two points per dataset, not a continuous curve; we did not train intermediate  $\delta$  values. (3) The cross-model comparison (Section 4) does not control for encoder capacity or training budget between F-CS-WAE and the four baselines. (4) We do not report a component-wise ablation of F-CS-WAE (e.g., removing the semantic variable, the class MMD term, or varying the prior geometry) beyond the  $\delta$  (per-class style MMD) axis, since we do not have completed runs for those variants. (5) The class-conditional style prior’s failure to fully transfer to CIFAR-10 (Section 6.7) is measured on one checkpoint per dataset; we have not tested whether a richer conditional (e.g., a per-class mixture rather than a single Gaussian) closes the gap.

## 8 Conclusion

Conditional style leakage is a failure mode in factorized generative models: global aggregate regularization of a style variable does not prevent it from becoming class-dependent, because marginal matching is necessary but not sufficient for conditional independence. We proved this gap, measured it with a three-metric diagnostic across five model families, and showed it is statistically real even when not visually obvious.

**Table 7:** CIFAR-10 generation under the same five sampling strategies,  $\delta=0$  checkpoint. Unlike MNIST, the parametric class-conditional strategies plateau well below 100%.

| Style sampling $z_s$     | Self-ACC     | Conf. | Div.         |
|--------------------------|--------------|-------|--------------|
| Global Gaussian          | 0.119        | 0.911 | 0.187        |
| Class mean               | 0.700        | 0.996 | 0.079        |
| Class diag., $\tau=0.50$ | 0.687        | 0.970 | 0.138        |
| Class diag., $\tau=0.25$ | 0.681        | 0.975 | 0.101        |
| Empirical style bank     | <b>0.959</b> | 0.988 | <b>0.241</b> |



**Figure 11:** CIFAR-10 analogue of Table 6. The class-conditional Gaussian strategies recommended for MNIST plateau near 70%; only resampling real encoded style codes (empirical bank) approaches MNIST-level self-accuracy.

Per-class style MMD cuts inter-class style separation by 67–77% and raises MNIST naive-sampling self-accuracy from 15% to 70% (100% combined with the post-hoc style prior); the clustering cost is negligible on CIFAR-10 but substantial on MNIST. Critically, the remedy that fully repairs MNIST generation only partially repairs CIFAR-10 generation, indicating that fixing the diagnostic finding does not automatically fix the underlying generative gap on more diverse data. Factorized generative models should verify class-invariance of style variables via conditional diagnostics, not marginal statistics alone, and should report where proposed remedies do and do not generalize.

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**Table 8:** Effect of per-class style MMD ( $\delta$ ), both real measurements.

| Dataset ( $\delta$ )    | ACC    | $\Delta_{\text{inter}}$ | Naive FID |
|-------------------------|--------|-------------------------|-----------|
| MNIST ( $\delta=0$ )    | 0.9915 | 6.27                    | 73.99     |
| MNIST ( $\delta=1$ )    | 0.8592 | 1.46                    | 43.36     |
| CIFAR-10 ( $\delta=0$ ) | 0.8149 | 4.19                    | 83.59     |
| CIFAR-10 ( $\delta=1$ ) | 0.8000 | 1.28                    | 74.42     |

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## A Architecture Details

The ResNet-18 backbone for  $32 \times 32$  inputs replaces the first  $7 \times 7$  convolution (stride 2) with a  $3 \times 3$  convolution (stride 1, padding 1) and removes the max-pool layer, yielding a

$4 \times 4$  spatial output with 512 channels. The shared FC layer maps  $512 \times 4 \times 4 = 8,192$  to 256 dimensions with SiLU. Semantic and style heads are separate linear layers from this 256-dimensional intermediate. The decoder uses three ResBlockUp stages (bilinear  $2 \times$  upsample,  $1 \times 1$  projection, GroupNorm+SiLU+Conv residual block), followed by a  $3 \times 3$  output convolution and sigmoid.

## B Training Hyperparameters

**Table 9:** Hyperparameters used in all F-CS-WAE experiments.

| Parameter                               | Value                            |
|-----------------------------------------|----------------------------------|
| Optimizer                               | Adam, lr = $10^{-3}$             |
| LR schedule                             | StepLR, step 100, $\gamma = 0.5$ |
| Batch size                              | 128                              |
| Total epochs                            | 300                              |
| Gradient clip (global norm)             | 1.0                              |
| Semantic dim $d_c$ / style dim $d_s$    | 64 / 128                         |
| Prior concentration $\rho_p$            | 0.7                              |
| EMA momentum $\tau$                     | 0.95                             |
| $\lambda_1 / \lambda_{\text{LPIPS}}$    | 1.0 / 0.1                        |
| $\alpha_f / \beta_f$ (class / agg. MMD) | 2.0 / 5.0                        |
| $\gamma_f$ (style)                      | 1.0                              |
| $\delta_f$ (per-class style, off/on)    | 0 / 1                            |
| $\eta_f$ (classifier)                   | 0.3                              |
| Style prior temperature $\tau_s$        | 0.25                             |

Baselines (ResNetAE, VAE, WAE-MMD, VaDE, CS-WAE) use a unified lightweight CNN protocol, 50 epochs, Adam lr  $10^{-3}$ , StepLR(30, 0.5); the cross-model diagnostic baselines of Section 4 use a separate lightweight 3-layer CNN encoder, 100 epochs (see Section 4 for the resulting comparability caveat).

## C Phase Schedule Details

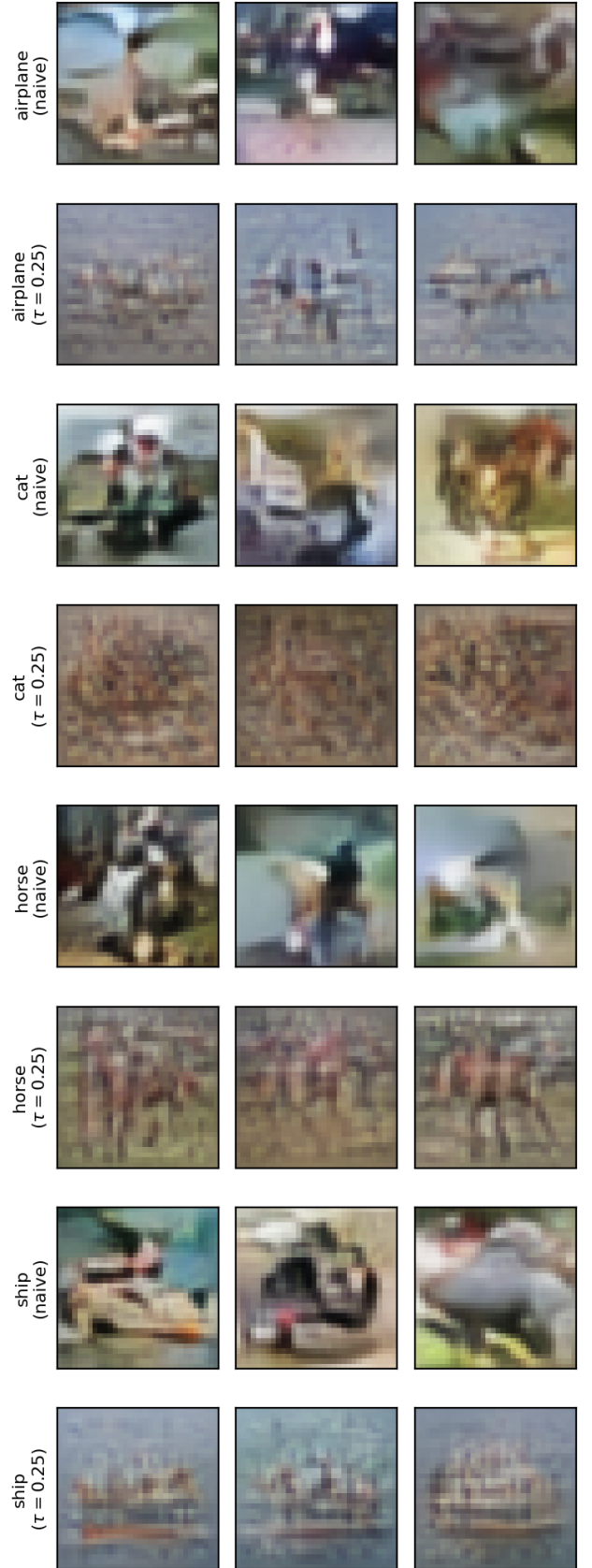
**Table 10:** Training phase boundaries and active loss terms; ramps are linear.

| Phase | Epochs  | Terms and ramp                                                                                                                                              |
|-------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | 0–49    | $\mathcal{L}_{\text{rec}}$ only                                                                                                                             |
| B     | 50–99   | $+\mathcal{L}_{\text{style}}(0 \rightarrow \gamma_f), \mathcal{L}_{\text{style-cl}}(0 \rightarrow \delta_f), \mathcal{L}_{\text{cls}}(0.1 \rightarrow 0.2)$ |
| C     | 100–199 | $+\mathcal{L}_{\text{class}}(0 \rightarrow \alpha_f/2), \mathcal{L}_{\text{agg}}(0 \rightarrow \beta_f/2), \mathcal{L}_{\text{cls}}(0.2 \rightarrow 0.3)$   |
| D     | 200–299 | $\mathcal{L}_{\text{class}}(\alpha_f/2 \rightarrow \alpha_f), \mathcal{L}_{\text{agg}}(\beta_f/2 \rightarrow \beta_f)$                                      |

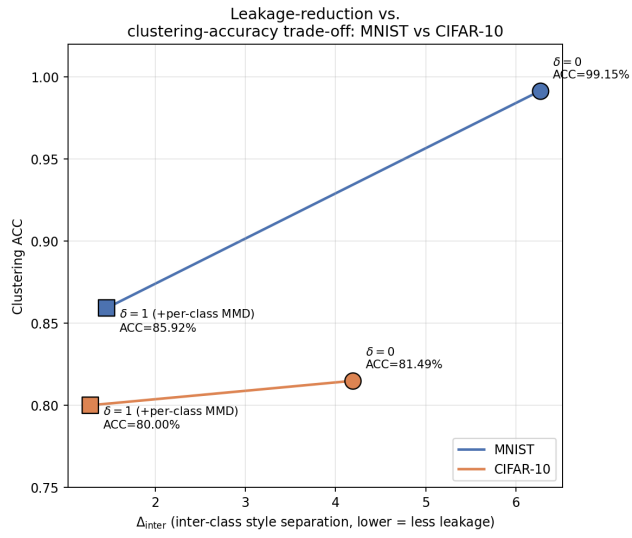
## D Proof of Proposition 1

Let  $K=10$ ,  $d_s=2$ ,  $q(z_s | y=k) = \mathcal{N}(\mu_k, \Sigma_k)$  with  $\mu_k = r(\cos(2\pi k/K), \sin(2\pi k/K))^\top$  for radius  $r>0$  and  $\Sigma_k = I - \frac{r^2}{1+r^2} \hat{\mu}_k \hat{\mu}_k^\top$ ,  $\hat{\mu}_k = \mu_k / \|\mu_k\|$ . The mixture mean is  $\frac{1}{K} \sum_k \mu_k = 0$  and the mixture covariance is  $I + \frac{r^2}{K} \sum_k \hat{\mu}_k \hat{\mu}_k^\top -$

$\frac{r^2}{K}I$ , which approaches  $I$  for this symmetric configuration. The inter-class mean separation is  $\Delta_{\text{inter}} \approx r\sqrt{2}$ , unbounded as  $r \rightarrow \infty$ , while the marginal stays  $\approx \mathcal{N}(0, I)$ : this shows  $\Delta_{\text{inter}}$  can be arbitrarily large even under an exactly matched marginal, proving the proposition.



**Figure 12:** CIFAR-10 naive (odd rows) vs. class-conditional  $\tau=0.25$  (even rows) samples, four classes. Diversity within a row collapses under  $\tau=0.25$ , and object identity is often not clearly recognizable by eye even where the auxiliary classifier scores it correctly (Section 7).



**Figure 13:** Clustering accuracy vs.  $\Delta_{\text{inter}}$  for the two measured configurations ( $\delta=0$ ,  $\delta=1$ ) per dataset. We show only these two points; we do not have intermediate  $\delta$  checkpoints and do not draw a continuous curve between them (Section 7.1).